$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi}} \int dx e^{\frac{-ipx}{\hbar}} \psi(x). \qquad (8.24)$$

This equation shows us exactly how to transform a given wave function in the position representation into the corresponding wave function in the momentum representation. What is it good for? Suppose the position wave function for some particle is known; however, the goal of your experiment is to measure the momentum, and you want to know the probability of observing momentum *p*. The procedure is to first calculate $\tilde{\psi}(p)$ by using Eq. 8.24 and then compute the probability

$$P(p) = \tilde{\psi}^*(p)\tilde{\psi}(p).$$

It's just as easy to go the other way. Suppose we know $\tilde{\psi}(p)$ and wish to recover $\psi(x)$. This time, we use Eq. 8.22 to resolve the identity. Here are the steps (notice that they look suspiciously similar to the earlier ones):

- First, use the definition of the position-representation wave function:

$$\psi(x) = \langle x|\Psi\rangle$$

- Now, insert the unit operator between the bra- and ket-vectors, in the form given in Eq. 8.22:

$$\psi(x) = \int dp \langle x|p\rangle \langle p|\Psi\rangle.$$

- The expression <*p*|Ψ> is just the wave function $\tilde{\psi}(p)$, and <*x*|*p*> is given to us by Eqs. 8.18. But this time, it's the first of the two equations.